

Day 6 — *Anurūpyeṇa*: Extending to $\times 12$, $\times 13$, ..., $\times 19$

Module: $\times 11$ (*Ekādhikena Pūrveṇa*) · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will apply Tirthaji's *Anurūpyeṇa* sub-sūtra ("proportionally") to multiply any 2-digit number by 12, 13, ..., 19, with at least 80% accuracy on a mixed worksheet.

Materials

- Whiteboard.
- `activities/activity-02-anurupyena.md` worksheet.
- Stopwatch.

Vocabulary introduced today

- *Anurūpyeṇa* (IAST: *anurūpyeṇa*; lit. "proportionally") — a sub-sūtra in Tirthaji's corpus. We use it as the name for the proportional scaling that extends the $\times 11$ pattern to $\times 12$, $\times 13$, ..., $\times 19$.

Lesson flow

Warm-up (5 min)

- Recall: state the $\times 11$ trick in one sentence.
- "*Outer digits unchanged; middle is the digit sum.*"

Core (18 min) — The scaling pattern

1. **Question.** "*What does $\times 12$ look like? Try 23×12 by long multiplication first.*"

$$23 \times 12 = 23 \times 10 + 23 \times 2 = 230 + 46 = 276.$$

2. **Derive.** "*Now distribute $(10a + b) \times 12 = (10a + b)(10 + 2)$:*

$$\begin{aligned} &= 100a + 10b + 20a + 2b \\ &= 100a + 10(b + 2a) + 2b \end{aligned}$$

Hmm — that's awkward. $2b$ in the units, $(b + 2a)$ in the tens. Let's organise it more cleanly.

3. **The Tirthaji pattern.** For N in $\{11, 12, \dots, 19\}$, write $N = 10 + k$ where $k \in \{1, 2, \dots, 9\}$. Then:

$$\begin{aligned} &(10a + b) \times (10 + k) \\ &= 100a + 10b \cdot k + 10ka + bk \\ &\quad \text{Wait — let's redo carefully:} \\ &= (10a + b) \times 10 + (10a + b) \times k \end{aligned}$$

$$\begin{aligned}
 &= 100a + 10b + 10ak + bk \\
 &= 100a + 10(b + ak) + bk
 \end{aligned}$$

Hmm — bk could be 2 digits. The pattern is: **scale each digit by k as you slide along.**

Easier form for students: For $\times N = \times(10 + k)$:

- Start with the leftmost digit.
- Multiply it by k , add the next digit, write the units of that sum (carry as needed).
- Continue right.
- The rightmost step: multiply the last digit by k .

This is the **Tirthaji form of the *Anurūpyeṇa* extension.**

4. **Worked example: 23×13 . ($k = 3$.)**

Set up: 0 2 3 (pad with 0 on left to track the carry-out)

Right to left:

- Units: last digit $\times k = 3 \times 3 = 9$. Write 9.
- Tens: middle digit $\times k + \text{next-right digit} = 2 \times 3 + 3 = 9$.

Write 9.

- Hundreds: leftmost digit $\times k + \text{next-right} = 0 \times 3 + 2 = 2$.

Wait — we want the LEFTMOST as just itself with the carry...

(Pause — this is getting confusing. Let's use a different presentation.)

5. **Cleaner presentation for $\times 12$ to $\times 19$.**

For $N \times 12$ of a 2-digit number ab : simply use long multiplication mentally, but in the order:

- units of answer = $b \times 2$ (carry if ≥ 10)
- tens of answer = $a \times 2 + b$ (plus carry; carry if ≥ 10)
- hundreds of answer = a (plus carry)

For $N \times 13$: replace $\times 2$ with $\times 3$ everywhere.

General: **For $\times(10 + k)$, multiply each digit by k and add the digit to its right (the “running scan”).**

6. **Examples (boxed):**

23×12 :

units: $3 \cdot 2 = 6$

tens: $2 \cdot 2 + 3 = 7$

hundreds: 2 (no carry)

→ 276 ✓

23×13 :

units: $3 \cdot 3 = 9$

tens: $2 \cdot 3 + 3 = 9$

hundreds: 2

→ 299 ✓

45 × 13:

units: $5 \cdot 3 = 15$ → write 5, carry 1

tens: $4 \cdot 3 + 5 + 1 = 18$ → write 8, carry 1

hundreds: $4 + 1 = 5$

→ 585 ✓

Check 45×13 : $45 \times 10 + 45 \times 3 = 450 + 135 = 585$. ✓

7. **The name.** “Tirthaji (1965) calls this proportional scaling **Anurūpyeṇa** — ‘proportionally.’ The $\times 11$ case is the special case $k = 1$, where the scaling is trivial.”

Activity (15 min) — Mixed drill

See activities/activity-02-anurupyena.md.

- 16 problems across $\times 12$, $\times 13$, $\times 14$, $\times 17$, $\times 19$.
- Paired self-check.

Wrap-up (2 min)

- “How does $\times 11$ fit into the $\times 12$ – $\times 19$ family?” (It’s the $k=1$ special case.)
- Preview Day 7: “What about $\times 23$? $\times 47$? When does the trick stop helping?”

Student-facing content

The $\times 12$ to $\times 19$ family — **Anurūpyeṇa**

For any $N = 10 + k$ (where k is 1 to 9), multiply a 2-digit number $ab \times N$ by: 1. Units = $b \times k$ (carry if ≥ 10). 2. Tens = $a \times k + b$ (plus any carry; carry if ≥ 10). 3. Hundreds = a (plus any carry).

When $k = 1$ (i.e. $\times 11$), this is the original “split and add” trick.

Tirthaji (1965) names this scaling pattern **Anurūpyeṇa** — “proportionally.”

Homework

- 12 problems mixing $\times 12$, $\times 13$, $\times 14$: see worksheet attached. Self-check with long multiplication on any 4.

Differentiation

- **Tier up:** Try 3-digit $\times 13$ problems. Use the right-to-left running-scan from Day 4 with the k -scaling from today.
- **Tier down:** Focus only on $\times 12$ today. Build fluency before adding higher k .

Teacher notes

- **Anurūpyeṇa** is the pattern-naming step. Some students will object: “This is just long multiplication done right-to-left.” They are correct. Honour the observation; reframe: “The win is doing it **MENTALLY** in one pass, with a clear rhythm.”

- The procedural complexity grows quickly with k . $\times 17$ and $\times 19$ are harder than $\times 11$ by a real margin. Calibrate the problem set accordingly.
- This day is genuinely harder than Days 1–5. If the class is shaky, slow down — skip the harder k values and revisit on Day 9.

Citation(s) used in this lesson

- Tirthaji (1965), sub-sūtra *Anurūpyeṇa*.
- *Microsoft Word - sutras1.pdf* — for the sub-sūtra list.